

4.4.a

$$H(t)^2 = \frac{8\pi G}{3c^2} \epsilon(t) - \frac{\kappa c^2}{R_0^2 a(t)^2} \quad (4.20)$$

$$\Omega(t) = \frac{\epsilon(t)}{\epsilon_t(t)} = \frac{8\pi G \epsilon(t)}{3c^2 H(t)^2} \leadsto \epsilon(t) = \frac{3c^2 H(t)^2}{8\pi G \Omega(t)}$$

$$\rightarrow \epsilon(t) = \frac{3c^2 H(t)^2}{8\pi G} \Omega(t)$$

$$H(t)^2 = \frac{\cancel{8\pi G}}{\cancel{3c^2}} \frac{3c^2 H(t)^2}{\cancel{8\pi G}} \Omega(t) - \frac{\kappa c^2}{R_0^2 a(t)^2}$$

$$H(t)^2 = H(t)^2 \Omega(t) - \frac{\kappa c^2}{R_0^2 a(t)^2}$$

$$\rightarrow \left[ H(t)^2 - H(t)^2 \Omega(t) = - \frac{\kappa c^2}{R_0^2 a(t)^2} \right] \div H(t)^2$$

$$1 - \Omega(t) = - \frac{\kappa c^2}{R_0^2 a(t)^2 H(t)^2}$$

$$\text{ii) } 4.34: \quad 1 - \Omega(t) = \frac{-kc^2}{R_0^2 a(t)^2 H(t)^2}$$

at  $t = t_0$  (Present)

$$\Omega_0 = \Omega(t_0)$$

$$a_0 = a(t_0)$$

$$H_0 = H(t_0)$$

$$1 - \Omega_0 = - \frac{kc^2}{R_0^2 H_0^2} \quad (4.35)$$

iii)

(4.35)

$$1 - \Omega_0 = - \frac{Kc^2}{R_0^2 H_0^2} \quad \times \frac{H_0}{c^2}$$

$$\rightarrow \frac{H_0^2}{c^2} (1 - \Omega_0) = - \frac{K}{R_0^2} \quad \times -1$$

$$\frac{H_0^2}{c^2} (\Omega_0 - 1) = \frac{K}{R_0^2}$$

iv)  
it says that  
any small deviation from  $\Omega \approx 1$  back in time  
would be much larger today  
i.e. if  $\Omega_0$  is close to 1  
(now)

then  $\Omega(t)$  at  $t \ll t_0$  is extremely close  
to 1

(Flatness problem)  
need inflation to fix

4.5)

i)  $p = w \epsilon$  (4.55)

$$w \approx \frac{v^2}{3c^2} \ll 1 \rightarrow w = 0 \quad (4.60)$$

Assuming  
nonrelativistic gas

Could be: cold dark matter  
baryons or neutral atoms

No pressure, energy density  $\epsilon \propto a^{-3}$

ii)

$$p_{\text{rel}} = \frac{1}{3} \epsilon_{\text{rel}} \rightarrow w = \frac{1}{3}$$

Photons, ultrarelativistic particles, early universe  
neutrinos.

Exerts pressure  $\epsilon \propto a^{-4}$  due to red shifting

iii)

$$w < -\frac{1}{3}$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2} (\rho + 3p) \rightarrow \frac{\ddot{a}}{a} > 0 \text{ if } w < -\frac{1}{3}$$

accelerated expansion  $\rightarrow$  dark energy or inflation,  
even before cosmological constant  
was introduced.

iv)

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2} \rho (1+3w)$$

$$w = -1$$

$$\frac{\ddot{a}}{a} = -\frac{4\pi G}{3c^2} \rho (1-3) = \frac{8\pi G}{3c^2}$$

$> 0 \rightarrow$  exponential  
and constant energy density

cosmological constant